

TUBEROUS SCLEROSIS COMPLEX / TSC1 / TSC2

Rare-Disease Health-Economic Case Study

Earlier identification and EEG surveillance in infants with tuberous sclerosis complex (TSC1/TSC2)

Conall R. Aque - Independent Researcher · October 2026
Independent applied-research project · Educational illustration · github.com/conallaque/genomelens

Illustrative HEOR analysis informed by published clinical literature. Numerical results are synthetic and do not represent a validated GenomeLens v1.11 model output. Only the clinical outcome proportions and the share of infants with a pre-seizure EEG trigger are taken from the cited trial. Every cost, resource-use and utility input is an independently chosen, rounded, illustrative value. None is an estimate of an actual price, cost or utility, and none is clinical or economic evidence.

1. Research question and population

Question. What could be the health-economic consequences of identifying an infant with a pathogenic TSC1 or TSC2 variant before the first seizure and offering serial EEG surveillance, with preventive vigabatrin when epileptiform activity appears, compared with usual clinical recognition and treatment after seizure onset? Epilepsy develops in 70-90% of children with tuberous sclerosis complex and is often resistant to medication [1].

Population. Infants aged 0-24 months with a pathogenic or likely pathogenic TSC1/TSC2 variant and no prior seizure, in a US health-care setting. The relevant incremental population is infants whose diagnosis would otherwise follow clinical presentation. Infants already recognized before seizures (for example through prenatal findings or family history) are surveilled under usual care and gain nothing incremental from a genomic result.

2. Strategies, perspective and horizon

Element	Specification
Intervention	Genomic identification, then serial EEG surveillance to age 24 months, with vigabatrin started when epileptiform EEG activity appears before seizures (preventive treatment), plus standard TSC follow-up [6].
Comparator	Usual clinical recognition, with vigabatrin or other treatment started after the first seizure.
Perspective	US health-care sector (direct medical costs). Caregiver, educational and productivity effects are excluded.
Horizon, discounting	10 years (birth to age 10); costs and QALYs discounted at 3% per year [4]. Trial evidence covers 24 months, so epilepsy outcomes from month 24 to year 10 are an extrapolation assumption.
Threshold	λ = \$100,000 per QALY in the base case; \$50,000 and \$150,000 in sensitivity analysis.

3. Published clinical evidence

Table 1. Two trials of preventive versus conventional vigabatrin in infants with TSC

	EPISTOP [1]	PREVeNT [2]
Design	Multicentre; 94 infants with TSC and no prior seizure, monthly video-EEG. Vigabatrin given preventively on pre-seizure epileptiform EEG, or conventionally after the first electrographic or clinical seizure, under the same monitoring. Randomized (6 sites) and open-label (4 sites) cohorts, pooled for the 24-month analysis.	Phase IIb multicentre, randomized, double-blind, placebo-controlled. Vigabatrin at first epileptiform EEG versus placebo until seizure onset, then open-label vigabatrin. Enrolled at ≤6 months; followed to 36 months.
Infants	54 of 94 (57%) developed epileptiform EEG abnormalities before seizures (27 randomized, 27 open-label).	84 enrolled; 56 randomized (29 vigabatrin, 27 placebo); 19 of 27 placebo infants moved to open-label vigabatrin, median 44 days after randomization.
Drug-resistant epilepsy, 24 months	28% vs 64% (preventive vs conventional; pooled odds ratio 0.23).	48% vs 52% (vigabatrin vs placebo; p = 0.47).
Infantile spasms	0% vs 40% (pooled).	Hazard ratio 0.26 (95% CI 0.10-0.71); comparison not prespecified.
Other	Time to first clinical seizure longer with preventive treatment (randomized cohort: 364 vs 124 days).	Bayley-III cognitive composite 80.9 vs 83.9 (p = 0.87).

The trials disagree on the outcome that carries most of the economic weight. EPISTOP found less drug-resistant epilepsy with preventive treatment, in a pooled comparison that includes a non-randomized cohort. PREVeNT, the placebo-controlled randomized trial, found no difference in drug-resistant epilepsy or cognition at 24 months, but fewer and later infantile spasms. Both are small, and PREVeNT reported that TSC1 and variants of uncertain significance, which may mark milder disease, were more common in the placebo arm. The evidence does not settle whether preventive treatment reduces drug-resistant epilepsy, so the analysis treats that as a scenario question.

4. Illustrative economic model

A short decision tree. Every surveilled infant incurs surveillance cost; only the share f who develop a pre-seizure EEG trigger receive preventive treatment and are assumed to benefit, and no benefit is assumed for the rest. Program costs are placed at year 1, and spasm costs and QALY losses at year 0.5. Per infant under surveillance:

$$\Delta C = d(1) \cdot [C_{\text{surv}} + f \cdot C_{\text{treat}}] - f \cdot \Delta p_{\text{IS}} \cdot c_{\text{IS}} \cdot d(0.5) - f \cdot \Delta p_{\text{DRE}} \cdot c_{\text{DRE}} \cdot A$$

$$\Delta Q = f \cdot \Delta p_{\text{IS}} \cdot u_{\text{IS}} \cdot \tau \cdot d(0.5) + f \cdot \Delta p_{\text{DRE}} \cdot u_{\text{DRE}} \cdot A$$

$$\text{NMB} = \lambda \cdot \Delta Q - \Delta C, \text{ ICER} = \Delta C / \Delta Q$$

where

$\Delta p_{\text{IS}}, \Delta p_{\text{DRE}}$: difference in event probability between usual care and surveillance (IS, infantile spasms; DRE, drug-resistant epilepsy);

$C_{\text{surv}}, C_{\text{treat}}$: undiscounted surveillance cost for every infant, and preventive-treatment cost for an infant who is treated (Table 2);

$c_{\text{IS}}, c_{\text{DRE}}$: cost of one infantile-spasms episode, and extra annual cost of drug-resistant epilepsy; $u_{\text{IS}}, u_{\text{DRE}}$: the corresponding utility decrements; τ : duration of the infantile-spasms decrement (0.5 year);

$d(t) = 1/(1.03)^t$; $A = \sum_{t=3}^{10} 1/(1.03)^t = 6.617$, the present-value factor for drug-resistant epilepsy from age 2 to 10 (end-of-year convention).

Table 2. Inputs (per infant under surveillance; 2025 US dollars)

Input	Value	Status
Share with pre-seizure epileptiform EEG, f	0.57 (54/94)	Published [1]
P(drug-resistant epilepsy): conventional arm (proxy for usual care) / preventive arm	0.64 / 0.28	Published [1]
P(infantile spasms): conventional arm (proxy for usual care) / preventive arm	0.40 / 0.00	Published [1]; zero events in a small arm
Additional EEG studies × cost; neurology visits × cost	12 × \$1,000; 8 × \$400	Illustrative
Additional vigabatrin months × cost; safety assessments × cost	6 × \$2,000; 3 × \$500	Illustrative (treated infants only)
Cost of managing one infantile-spasms episode	\$20,000	Illustrative
Extra annual cost of drug-resistant epilepsy, ages 2-10	\$4,000	Illustrative
Utility decrement: drug-resistant epilepsy (per year); infantile spasms (0.5 year)	0.10; 0.30	Illustrative

5. Results

Table 3. Incremental results per infant under surveillance (λ = \$100,000 per QALY)

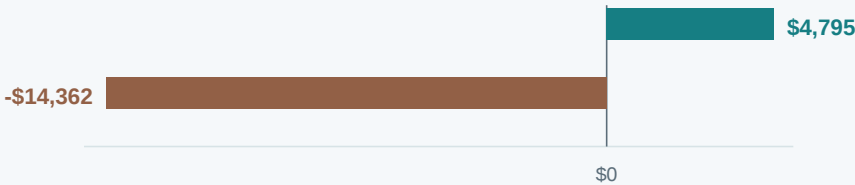
	Base case: EPISTOP-informed effects	Scenario B: no reduction in drug-resistant epilepsy
Surveillance and treatment cost (present value)	\$22,287	\$22,287
Infantile-spasms cost avoided	-\$4,528	-\$4,528
Drug-resistant-epilepsy cost avoided	-\$5,474	\$0
Incremental cost, ΔC	\$12,285	\$17,758
QALYs gained: infantile spasms; drug-resistant epilepsy	0.0340; 0.1368	0.0340; 0.0000
Incremental QALYs, ΔQ	0.1708	0.0340
ICER (\$ per QALY)	\$71,924	\$522,884
Net monetary benefit	\$4,795	-\$14,362

Net monetary benefit by effect scenario

Synthetic results per infant; λ = \$100,000 per QALY

EPISTOP-informed

No DRE reduction



In the base case, surveillance costs \$12,285 more per infant and yields 0.1708 QALYs (ICER \$71,924 per QALY). NMB is positive because each QALY is valued at λ , but the strategy is not cost-saving: ΔC is positive. Per 1,000 infants this is about \$12.3 million of added spending for 171 QALYs, an affordability question that NMB does not answer. At λ = \$50,000 NMB is negative. If preventive treatment does not reduce drug-resistant epilepsy (Scenario B), NMB is -\$14,362 and the ICER exceeds \$500,000 per QALY: the conclusion reverses with the evidence one believes.

6. One-way sensitivity analysis

Table 4. Net monetary benefit (\$ per infant) when each input is varied alone (eight largest of 14 inputs tested)

Input	Base	Setting A	NMB at A	Setting B	NMB at B
Time horizon	10 y	5 y	-\$6,643	18 y	\$19,919
P(drug-resistant epilepsy), preventive arm	0.28	0.64	-\$14,362	0.15	\$11,713
Willingness to pay per QALY	\$100,000	\$50,000	-\$3,745	\$150,000	\$13,336
Utility decrement, drug-resistant epilepsy (per year)	0.10	0.05	-\$2,047	0.15	\$11,637
Share of infants with a pre-seizure EEG trigger	0.57	0.40	-\$1,143	0.75	\$10,770
Cost per EEG study	\$1,000	\$1,500	-\$1,030	\$500	\$10,621
Additional vigabatrin months per treated infant	6	12	-\$1,897	3	\$8,142
Extra annual cost of drug-resistant epilepsy	\$4,000	\$2,000	\$2,059	\$8,000	\$10,269

NMB is below zero at one end of the range for 7 of the 14 inputs varied. A threshold analysis is more informative than any single row: holding the other inputs at base values, NMB is positive only if preventive treatment lowers the probability of drug-resistant epilepsy by at least about 27 percentage points. EPISTOP reports a 36-point difference; PREVeNT reports roughly none.

7. Probabilistic uncertainty

A probabilistic sensitivity analysis would place a distribution on every input (for example beta distributions on the clinical proportions, informed by the published event counts, and gamma distributions on costs), propagate them by Monte Carlo simulation, and report the distribution of NMB and the probability of cost-effectiveness across λ [3]. This illustration does not report a probability of cost-effectiveness. The economic inputs are illustrative, so any distribution placed on them would encode an arbitrary choice. The dominant uncertainty is also structural, not sampling variation within one study: two trials disagree about the effect on drug-resistant epilepsy. A single probabilistic analysis would hide that disagreement behind apparent precision. Even the published proportions are imprecise. Zero infantile-spasm events among about 25 infants is a point estimate, and by the rule of three the upper 95% bound is near 12%.

8. Interpretation, limitations and conclusion

Interpretation. Under EPISTOP-informed effects and these illustrative inputs, early surveillance buys health at an ICER below \$100,000 per QALY while increasing spending; under a PREVeNT-like null effect on drug-resistant epilepsy it does not. The result is most sensitive to the time horizon, the size of the effect on drug-resistant epilepsy, the threshold and the utility decrement, that is, to extrapolation and to contested evidence. It also turns negative if EEG studies cost 50% more or vigabatrin exposure doubles.

Limitations. Evidence: two small trials that conflict, one partly non-randomized. Both compared treatment timing within infants who were already diagnosed and EEG-monitored, so applying EPISTOP's conventional arm to unidentified infants under usual care is an assumption whose direction is not established. Population: the analysis assumes infants who would otherwise be recognized only after seizures; if most are already recognized earlier, the incremental population is small. Structure: benefit is applied only to infants with an EEG trigger; outcomes after month 24 are extrapolated; neurodevelopmental outcomes, mortality, other TSC manifestations and treatments, caregiver effects, vigabatrin harms (including visual-field loss) and false-positive findings are not modeled. Inputs: every cost and utility is illustrative. A model that runs is a research scenario, not a validated outcome.

Conclusion. This analysis shows how published clinical evidence, explicit assumptions and standard decision-analytic arithmetic combine into ΔC , ΔQ , ICER and NMB; why a positive NMB is not a cost saving; and how a conclusion can depend on which of two conflicting trials is believed. It does not establish that early surveillance is, or is not, cost-effective, and it is not evidence about any patient, price or policy. It follows the structure of the CHEERS 2022 reporting guidance where applicable [5] without claiming compliance.

References

- [1] Kotulska K, Kwiatkowski DJ, Curatolo P, et al. Prevention of epilepsy in infants with tuberous sclerosis complex in the EPISTOP trial. *Ann Neurol* 2021;89(2):304-314. doi:10.1002/ana.25956
- [2] Bebin EM, Peters JM, Porter BE, et al. Early treatment with vigabatrin does not decrease focal seizures or improve cognition in tuberous sclerosis complex: the PREVeNT trial. *Ann Neurol* 2024;95(1):15-26. doi:10.1002/ana.26778
- [3] Briggs AH, Weinstein MC, Fenwick EAL, et al. Model parameter estimation and uncertainty analysis: a report of the ISPOR-SMDM Modeling Good Research Practices Task Force Working Group-6. *Med Decis Making* 2012;32(5):722-732. doi:10.1177/0272989X12458348
- [4] Sanders GD, Neumann PJ, Basu A, et al. Recommendations for conduct, methodological practices, and reporting of cost-effectiveness analyses: Second Panel on Cost-Effectiveness in Health and Medicine. *JAMA* 2016;316(10):1093-1103. doi:10.1001/jama.2016.12195
- [5] Husereau D, Drummond M, Augustovski F, et al. Consolidated Health Economic Evaluation Reporting Standards 2022 (CHEERS 2022) statement. *Clin Ther* 2022;44(2):158-168. doi:10.1016/j.clinthera.2022.01.011
- [6] Northrup H, Aronow ME, Bebin EM, et al. Updated international tuberous sclerosis complex diagnostic criteria and surveillance and management recommendations. *Pediatr Neurol* 2021;123:50-66. doi:10.1016/j.pediatrneurol.2021.07.011